

		For a graph of S vs V , gradient of the line is radius r .
17	C	Energy dissipated across resistor, $E = Pt = IVt = (0.5)(1.5)(2 \times 60) = 90 \text{ J}$ Energy dissipated across internal resistor, $240 - 90 = 150 = I^2 R t = (0.5)^2 (R)(2 \times 60) = 90 \text{ J}$ $r = 3 \text{ } \Omega$
18	C	